

Goals of Software Testing

Why do we test? Three categories of goals every tester and manager keeps in mind.

— Software Testing Module 1

Topics's Agenda

Three categories of goals, and one real-world case study

01



Immediate (short-term) goals — bug discovery

02



A real product that suffered from a short-term-goal failure

03



Long-term goals — reliability, quality, satisfaction, risk

04



Post-implementation goals — maintenance, prevention, process

Three Categories of Testing Goals

Every goal of software testing falls on one of three timelines.



Immediate Goals

During testing

- Bug discovery



Long-term Goals

Across the product's life

- Reliability
- Quality
- Customer satisfaction
- Risk management



Post-Implementation Goals

After deployment

- Reduced maintenance cost
- Improved testing process
- Bug prevention

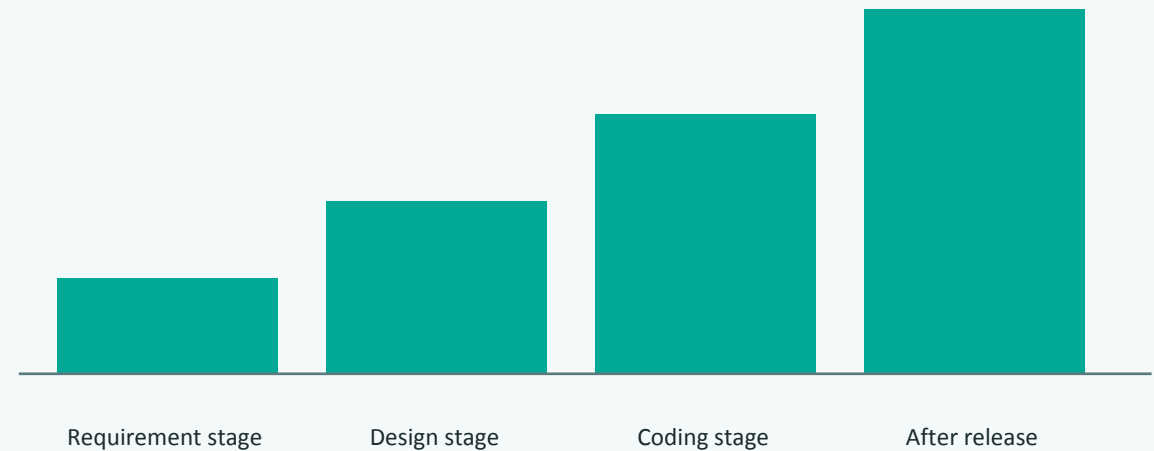
Immediate Goals: Bug Discovery



The immediate goal of testing is to find errors — at any stage of software development.

The earlier a bug is found, the cheaper and easier it is to fix — and the higher the overall success rate of the testing effort.

Cost of fixing a bug rises the later it's found



Real-World Case: When Bug Discovery Fails



Knight Capital Group · August 1, 2012

New trading software was deployed to production with an untested, leftover code path still active.

45

minutes for the glitch to run unnoticed

\$460M

lost before the system was shut down

1

day before the firm needed a bailout to survive

The lesson: this was not a subtle logic error. It was old, unused code that testing never re-verified before going live. A disciplined short-term goal of bug discovery — through proper regression and deployment testing — would very likely have caught it before a single trade was placed.

Long-Term Goals

What testing builds across the entire life of the product.



Reliability

The software consistently performs its intended function without failure, release after release.



Quality

The product meets defined standards of correctness, robustness, and craftsmanship.



Customer satisfaction

Users trust the product enough to keep using it, recommend it, and stay loyal.

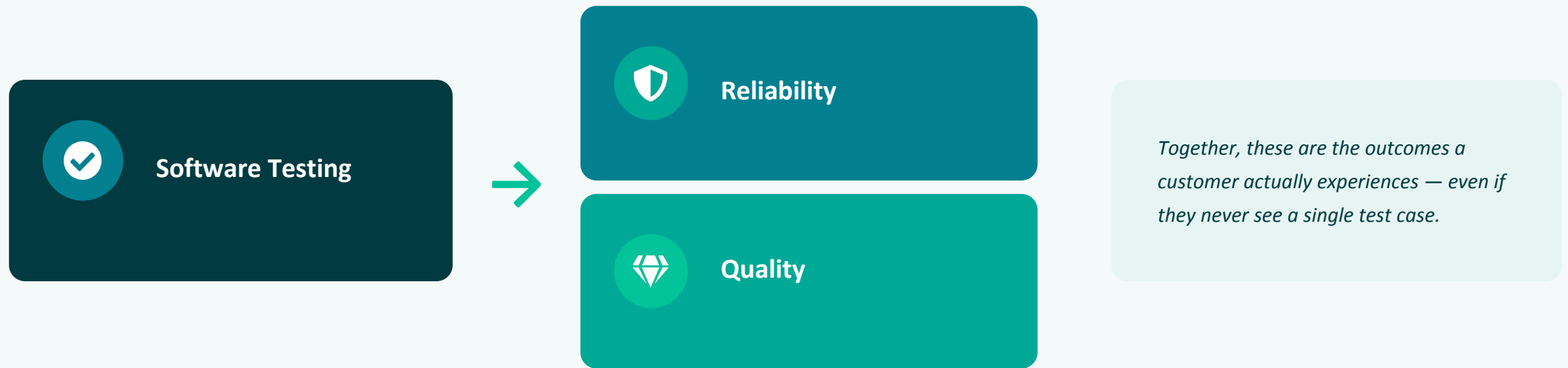


Risk management

Potential failures are identified and controlled before they can cause serious harm.

Testing Produces Reliability & Quality

A simple but powerful relationship



Reliability = it keeps working correctly over time. Quality = it meets the standard the user expected.

Post-Implementation Goals

What good testing leaves behind, after the product goes live.



Reduced maintenance cost

Fewer defects reach production, so support and patching consume less budget.



Improved testing process

Lessons from this release refine test plans, tools, and coverage for the next one.



Bug prevention

Root-cause analysis stops entire classes of defects from recurring in future code.

Key Takeaways

- ✓ Testing goals fall into three timelines: immediate, long-term, and post-implementation.
- ✓ The core immediate goal is bug discovery — and finding bugs earlier is always cheaper.
- ✓ Knight Capital lost \$460M in 45 minutes in 2012 from an untested, leftover code path.
- ✓ Long-term goals — reliability, quality, satisfaction, risk management — reinforce each other.
- ✓ Post-implementation goals turn each release into lower cost and fewer future bugs.